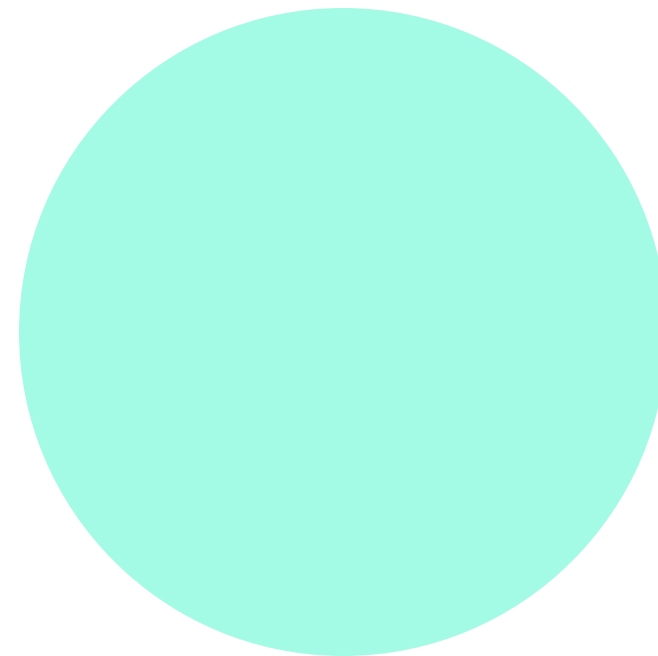


Inflation prediction in emerging economies: Machine learning and FX reserves integration for enhanced forecasting



NAME - Tanishq soni

Research Paper

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Introduction

What is Inflation Forecasting?

Inflation forecasting refers to the process of predicting future inflation rates using statistical, econometric, or machine learning models. It involves analyzing historical economic data and key indicators to estimate how the general price level of goods and services will change over time.

Why is Inflation Forecasting Important?

- **Policy Planning:** Central banks rely on inflation forecasts to set interest rates and manage money supply, ensuring price stability and economic growth.
- **Investment Decisions:** Accurate forecasts help investors and businesses make informed decisions about spending, saving, and investing.
- **Budgeting & Financial Planning:** Governments, firms, and households use forecasts to plan budgets, wages, and contracts.
- **Crisis Management:** Helps in identifying inflation shocks early, so corrective measures can be taken.

MOTIVATION

1. Rising Inflation Volatility in Emerging Economies

- Countries like India face frequent inflation shocks due to oil prices, currency fluctuations, and external debt.
- Accurate inflation forecasting is critical for monetary policy, investment decisions, and price stability.

2. Limitations of Traditional Models

- Widely used models (ARMA, ARIMA, Regression) assume linear relationships, struggle with non-stationary data, and exclude key real-world variables.
- These models often fail to predict inflation trends accurately, especially during economic shocks.

3. Underutilized Role of Foreign Exchange Reserves

- FX Reserves impact exchange rates, confidence, and inflation — yet are rarely included in forecasting models.
- Their stabilizing role is crucial in developing economies.

4. Machine Learning as a Game Changer

- ML models can capture non-linear relationships, learn from large datasets, and adjust to complex macroeconomic patterns.
- Opportunity to test whether ML outperforms traditional approaches in forecasting inflation more accurately.

5. Filling a Research Gap

- No comprehensive study has evaluated both ML and FX Reserves together for inflation prediction in emerging economies.
- This study seeks to bridge that gap — using India as a case study.

Objective

Two key goals:

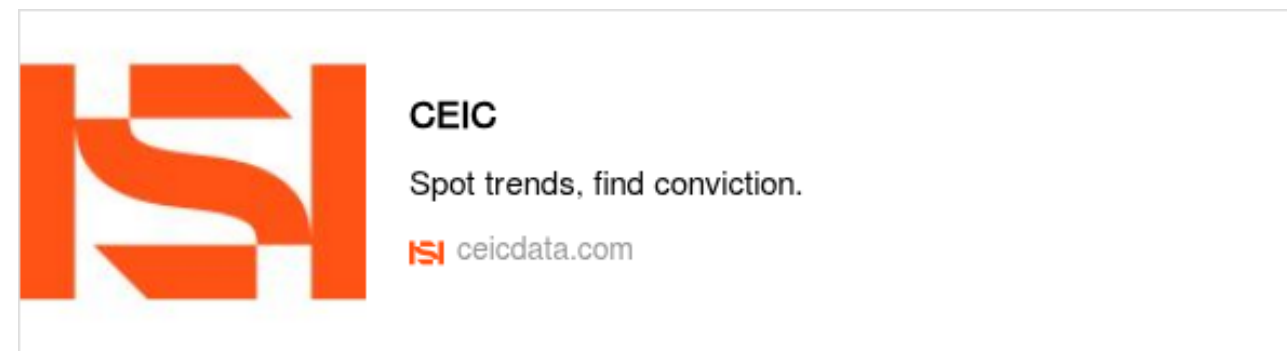
- Compare traditional and ML models for inflation prediction
- Test the impact of FX Reserves on forecast accuracy

Dataset

- Country: INDIA (2012–2023, monthly)
- Variables:

Target: Inflation

Features: IntRate, M2, Export, Import, IP, XR, FXReserves



Model structure

The following equations broadly explain the structure of our empirical models, which will be estimated using a combination of traditional and ML forecasting approaches to achieve the proposed study objective.

Inflation = f (IntRate, M2, Export,Import,IP) (Model 1)

Inflation = f (IntRate, M2, Export,Import,IP, XR) (Model 2)

Inflation = f (IntRate, M2, Export,Import,IR, XR, FXReserves) (Model 3)

where:

- IntRate is interest rate,
- M2 is money supply,
- IP is industrial production,
- XR is exchange rate
- FXReserves is foreign exchange reserves

Model 1 is the initial model of the study. Building upon the foundations established by the initial model, we extend our analytical framework by incorporating XR in model 2, and FXReserves in model 3.

Multiple regression

Multiple regression is a statistical technique that examines the relationship between one dependent variable and two or more independent variables. It aims to predict the value of the dependent variable based on the values of the independent variables.

$$Y_t = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_n X_n + \epsilon_t$$

- **Y_t**: The dependent variable (Inflation, denoted as INF)
- **X₁, X₂, ..., X_n**: Independent variables (e.g., interest rates, FXReserves, money supply)
- **β₀**: Intercept
- **β₁, β₂, ..., β_n**: Slope coefficients showing the effect of each variable
- **ε_t**: Error term (difference between actual and predicted values)

The model fits a straight line (or hyperplane in multiple dimensions) through the data points by finding the best coefficients β that minimize the error between the predicted and actual values.

This is achieved using Ordinary Least Squares (OLS), a method that minimizes the sum of squared residuals:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Limitations of Multiple Regression

1. Assumes Linear Relationships : Multiple regression cannot capture these complexities, leading to underfitting in dynamic economies like India.

2. Limited Predictive Power

As seen in the test phase of the graphs (green vs. purple), predictions deviate notably from actual values—especially in Model 1.

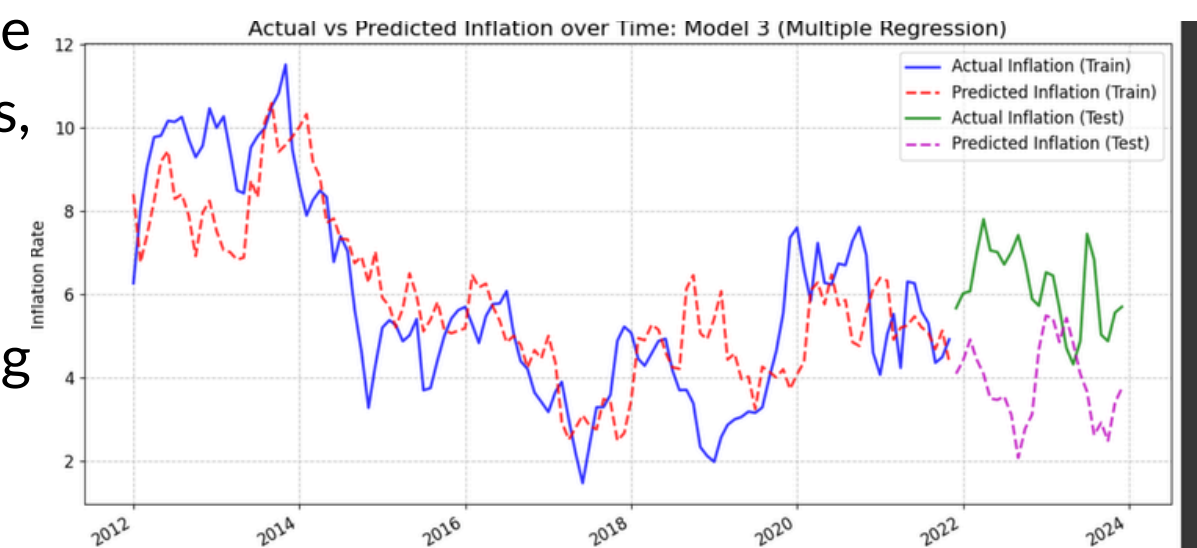
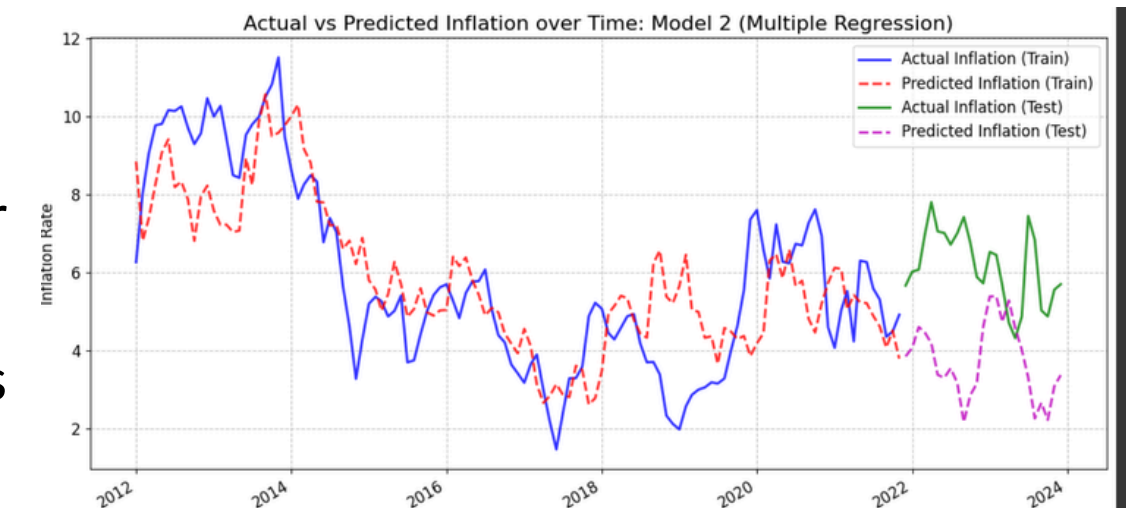
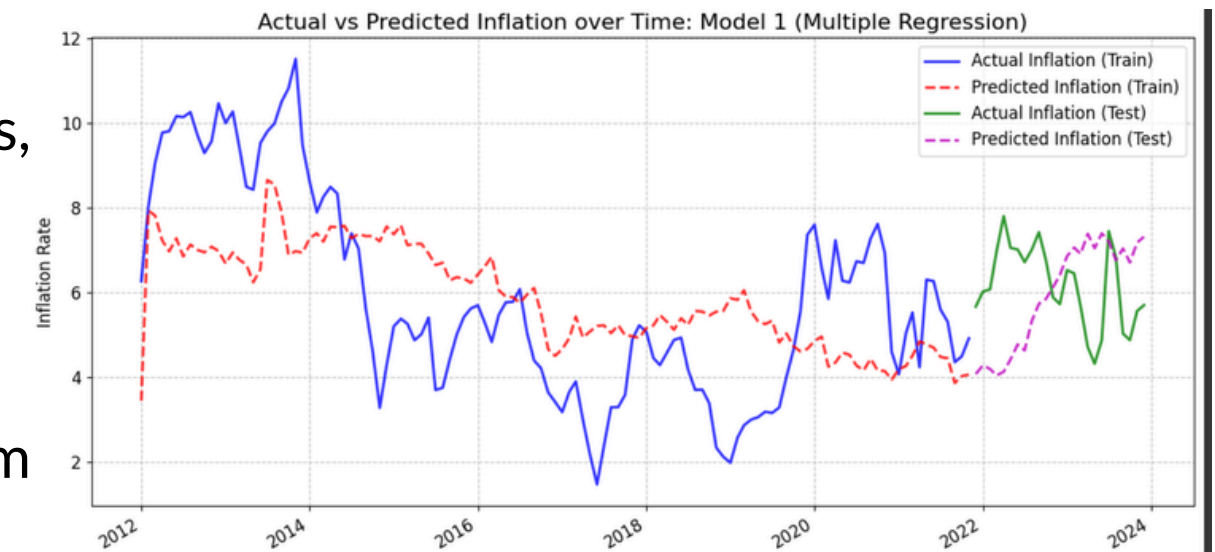
Suggests poor generalization to unseen data.

3. Insensitive to Structural Shocks :Events like demonetization (2016), COVID-19 (2020), or global oil shocks aren't captured well by static coefficients.

4. Collinearity Risk: Economic variables like exchange rate, money supply, and FX reserves may be interrelated, leading to multicollinearity that distorts coefficient estimates.

5. Stationarity & Autocorrelation Issues: Time series regression assumes stable mean/variance and no autocorrelation. Inflation data often violate these assumptions, affecting reliability.

6. Over-Reliance on Historical Patterns: Multiple regression models are backward-looking and can't adapt to regime shifts in monetary policy or global markets.



ARMA Model (Autoregressive Moving Average)

- A time series forecasting model that combines:
 - AR (Autoregressive): uses lagged values of inflation
 - MA (Moving Average): uses lagged forecast errors
- Best suited for stationary data (constant mean and variance over time)



Mathematical Form (ARMA(p,q))

$$\text{INF}_t = \beta_0 + \beta_1 \text{INF}_{t-1} + \cdots + \beta_p \text{INF}_{t-p} + \alpha_1 \epsilon_{t-1} + \cdots + \alpha_q \epsilon_{t-q} + \epsilon_t$$

Where:

- $\beta_1, \dots, \beta_p$: AR coefficients
- $\alpha_1, \dots, \alpha_q$: MA coefficients
- ϵ_t : error term

LIMITATIONS

Fixed Memory:

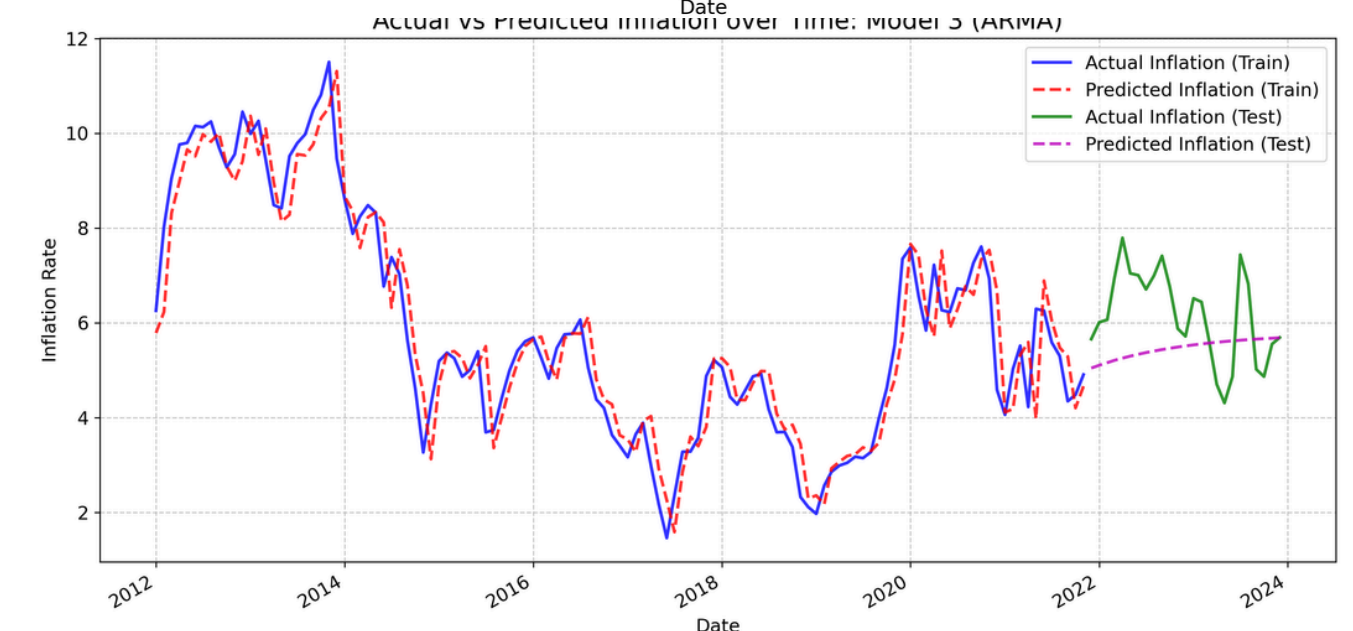
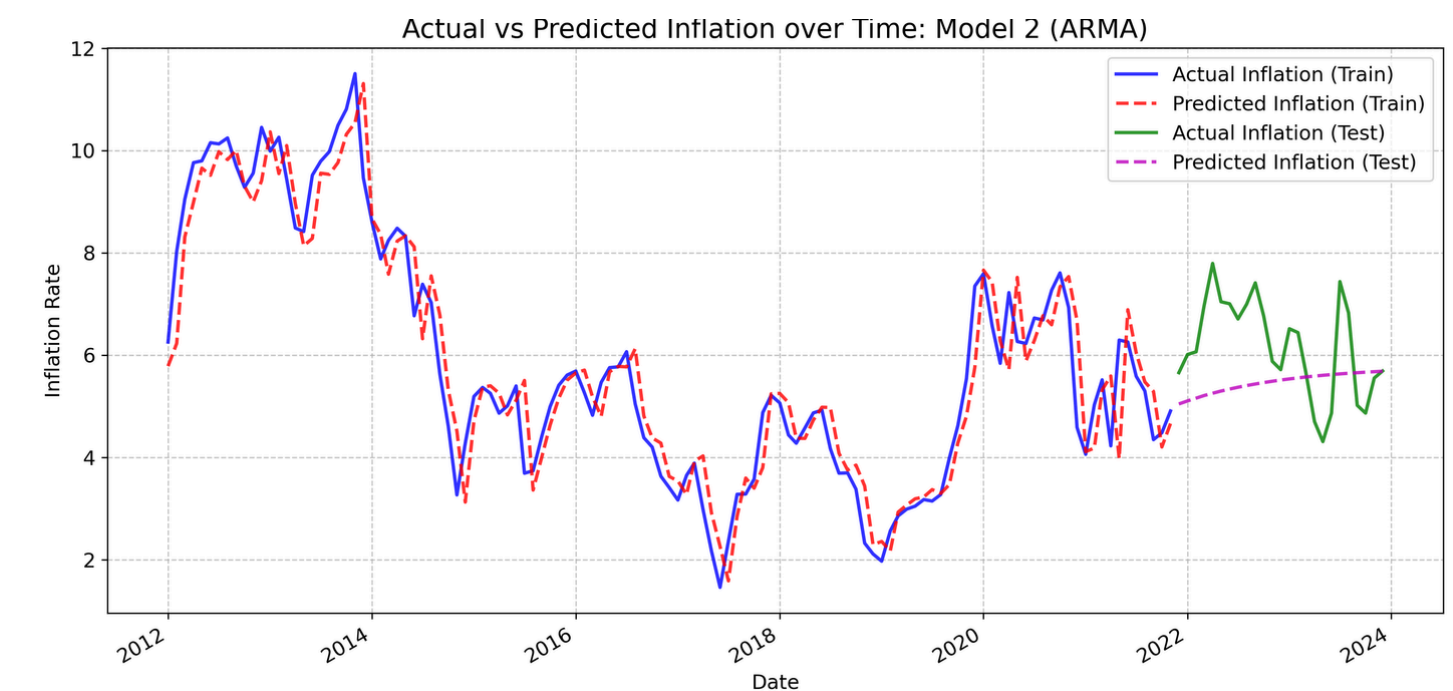
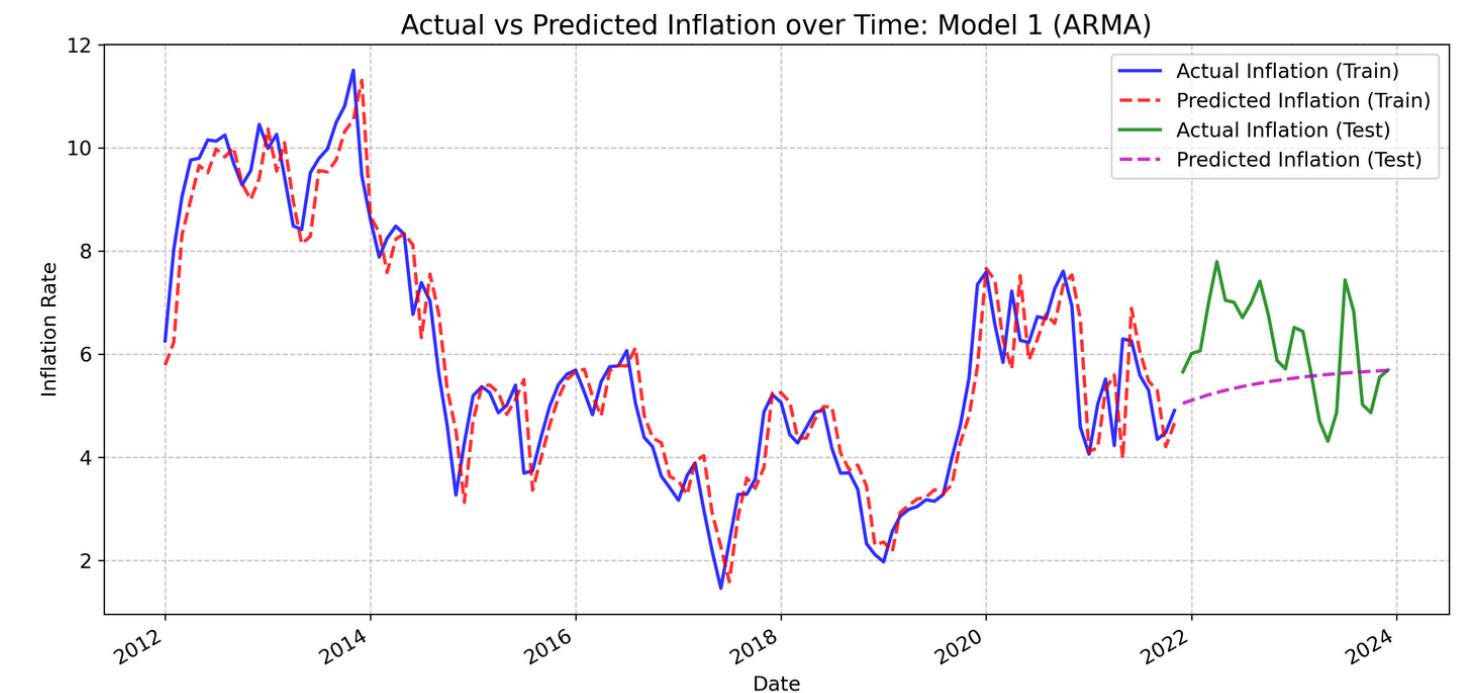
Considers only a fixed number of past observations and forecast errors.

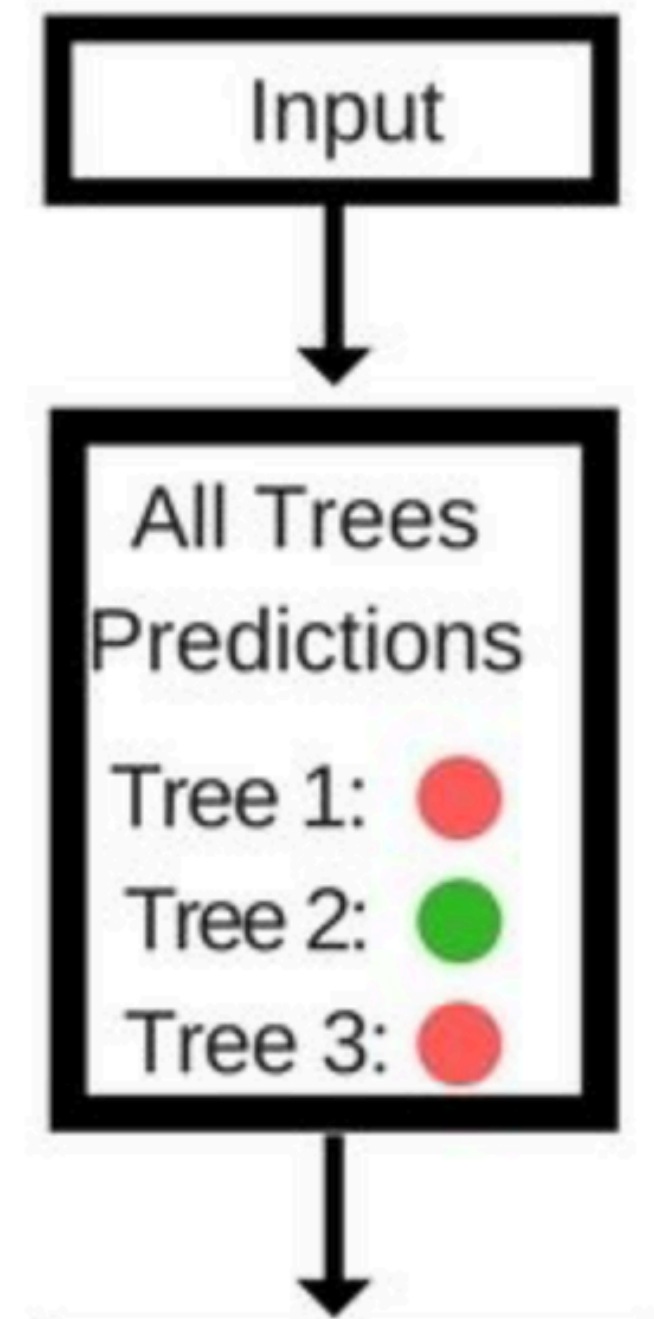
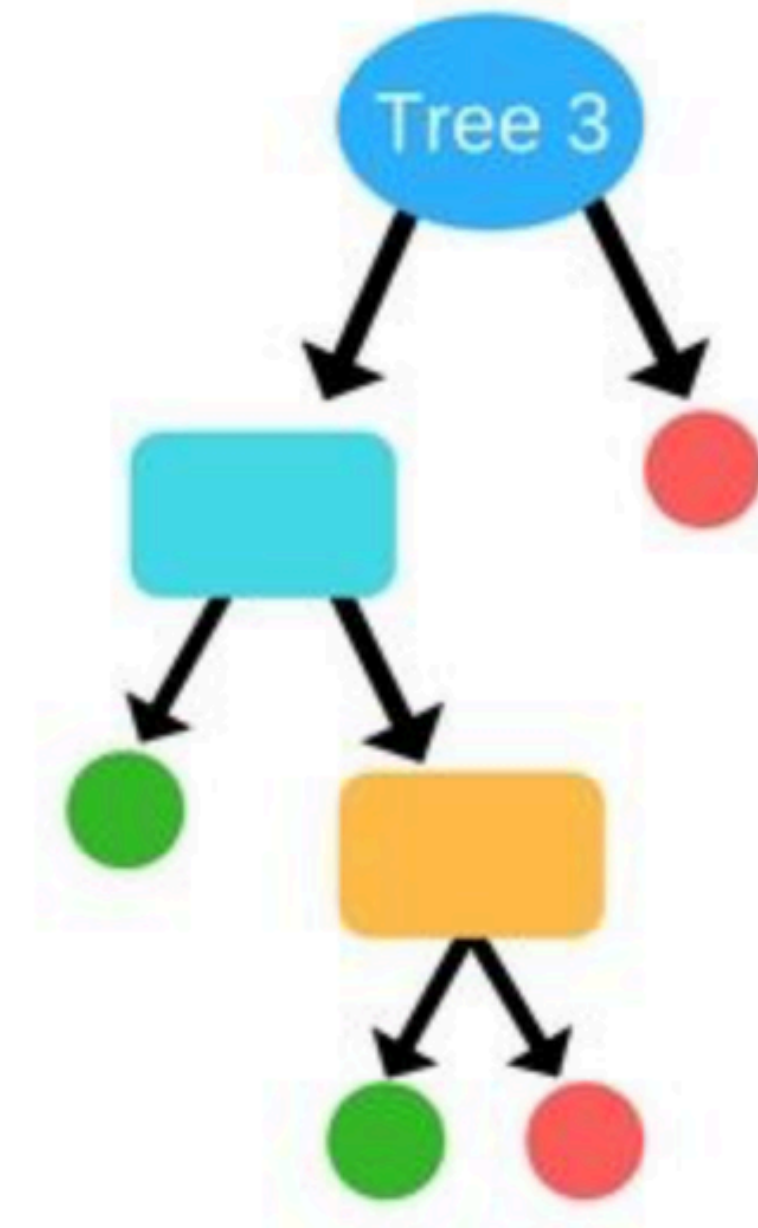
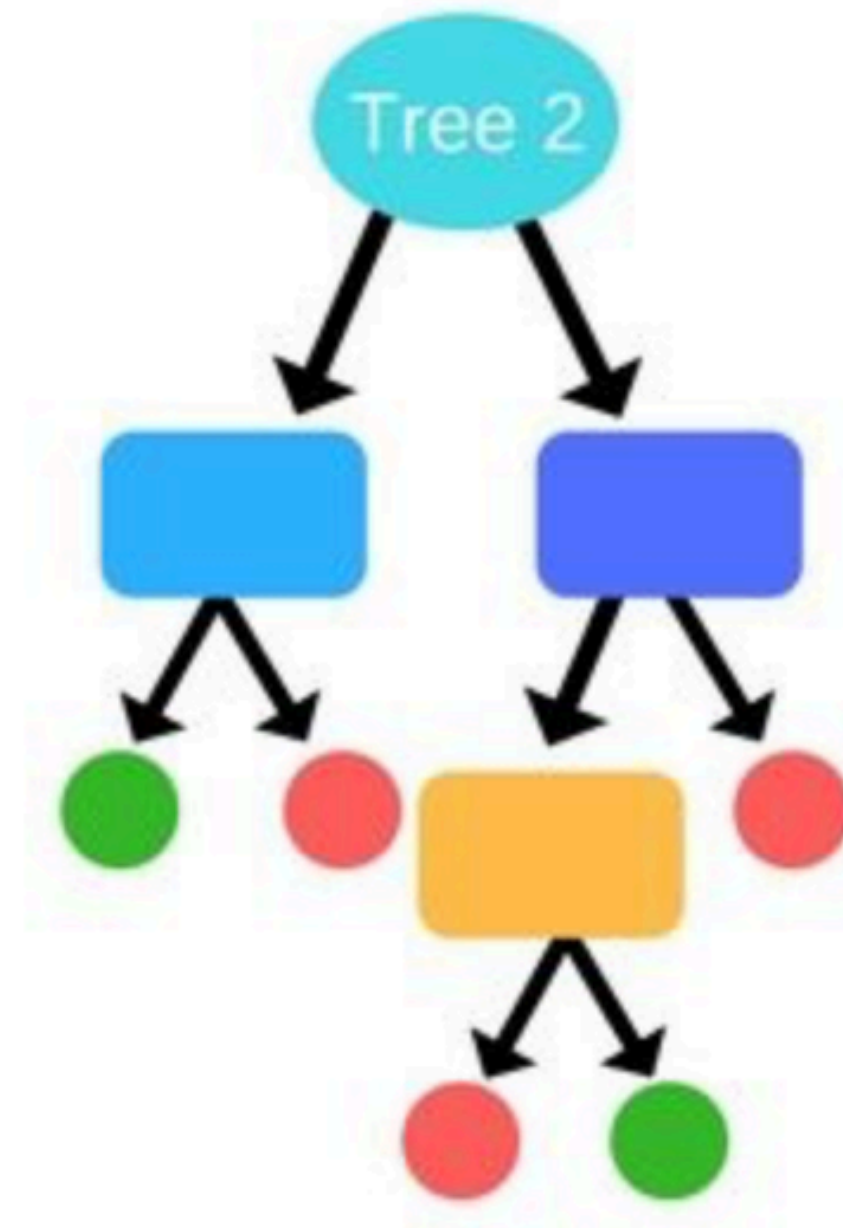
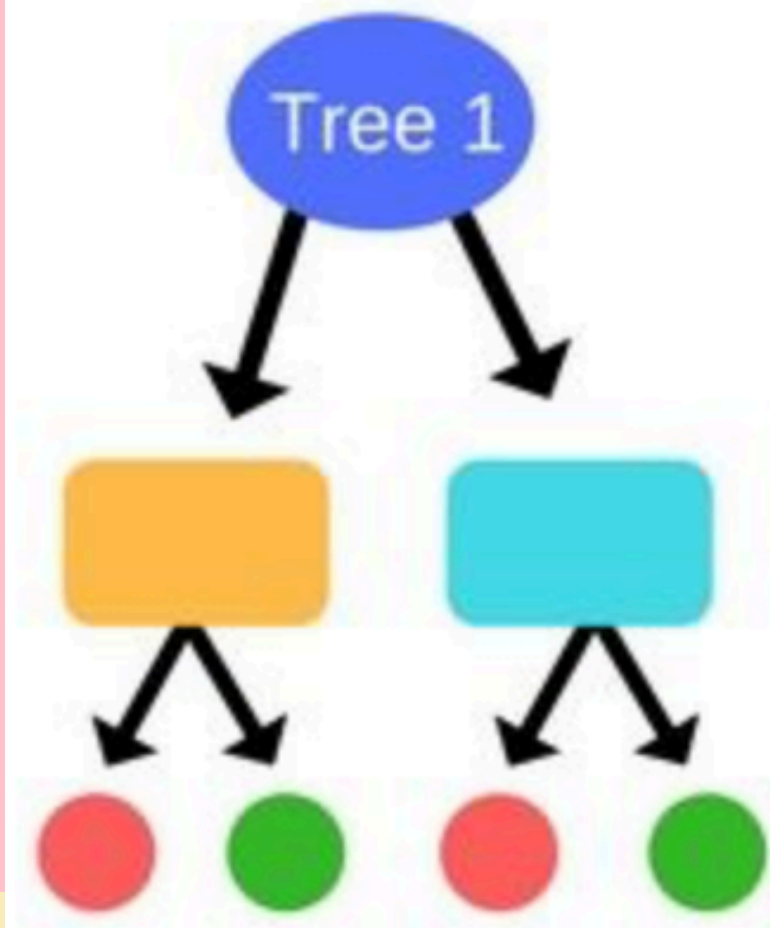
➤ Less effective for long-term forecasting.

Assumes Constant Statistical Properties:

Requires mean, variance, and covariance to remain constant over time.

➤ Leads to biased estimates if the series is non-stationary.



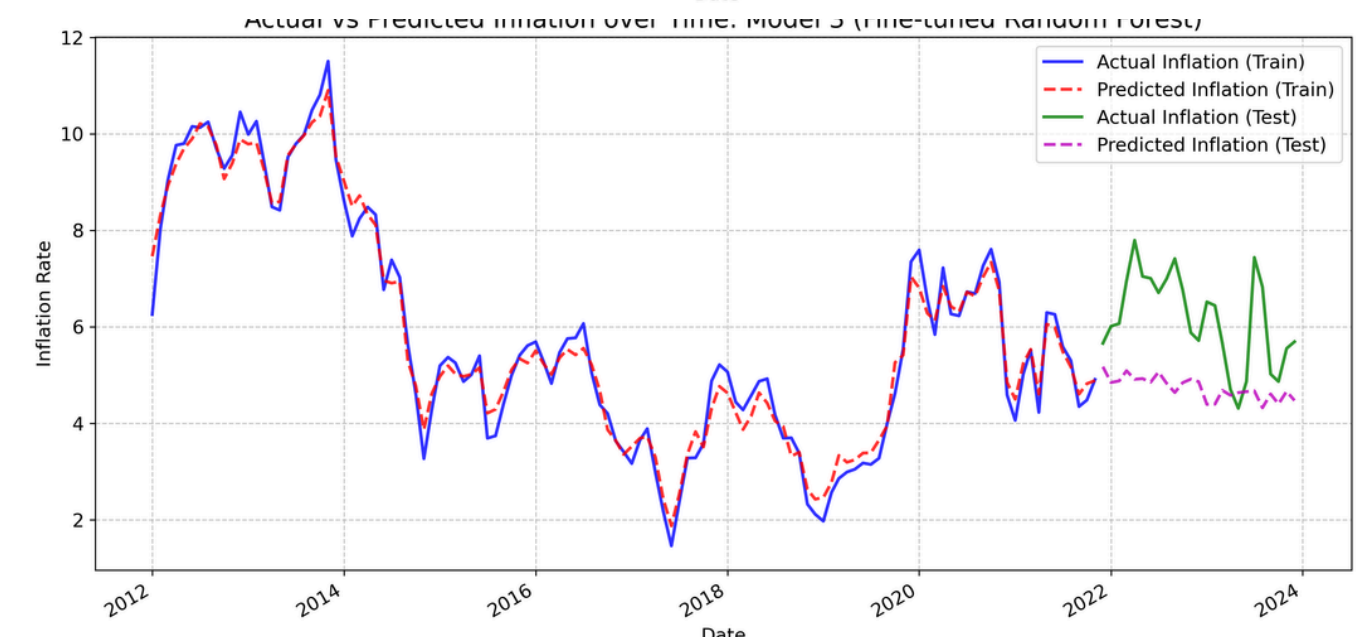
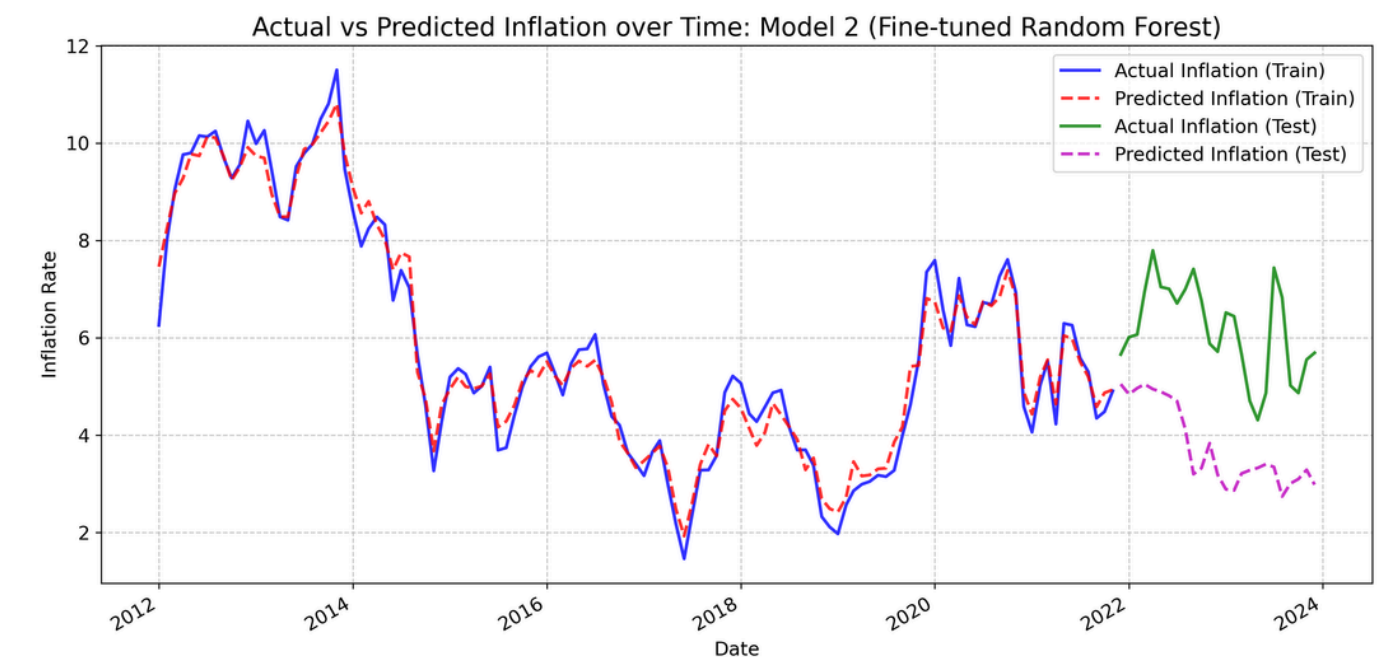
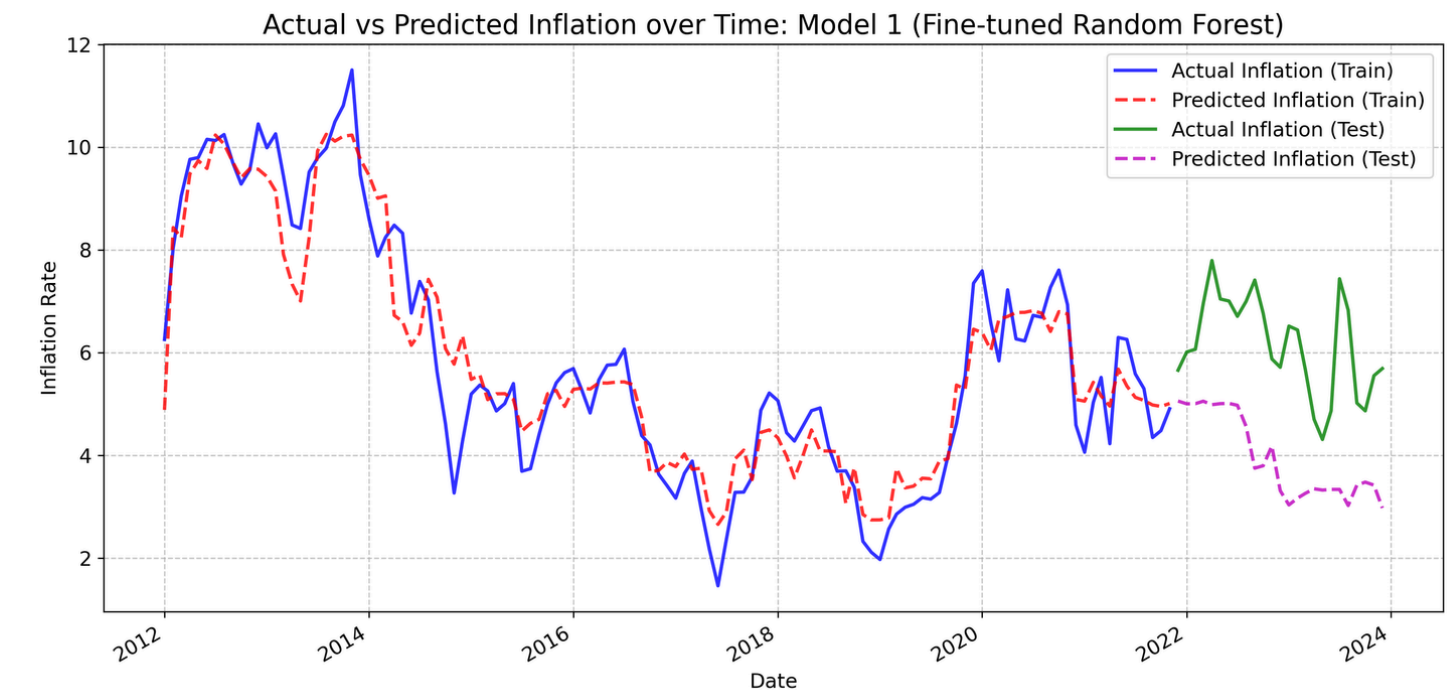


Random Forest: Power of Many Trees

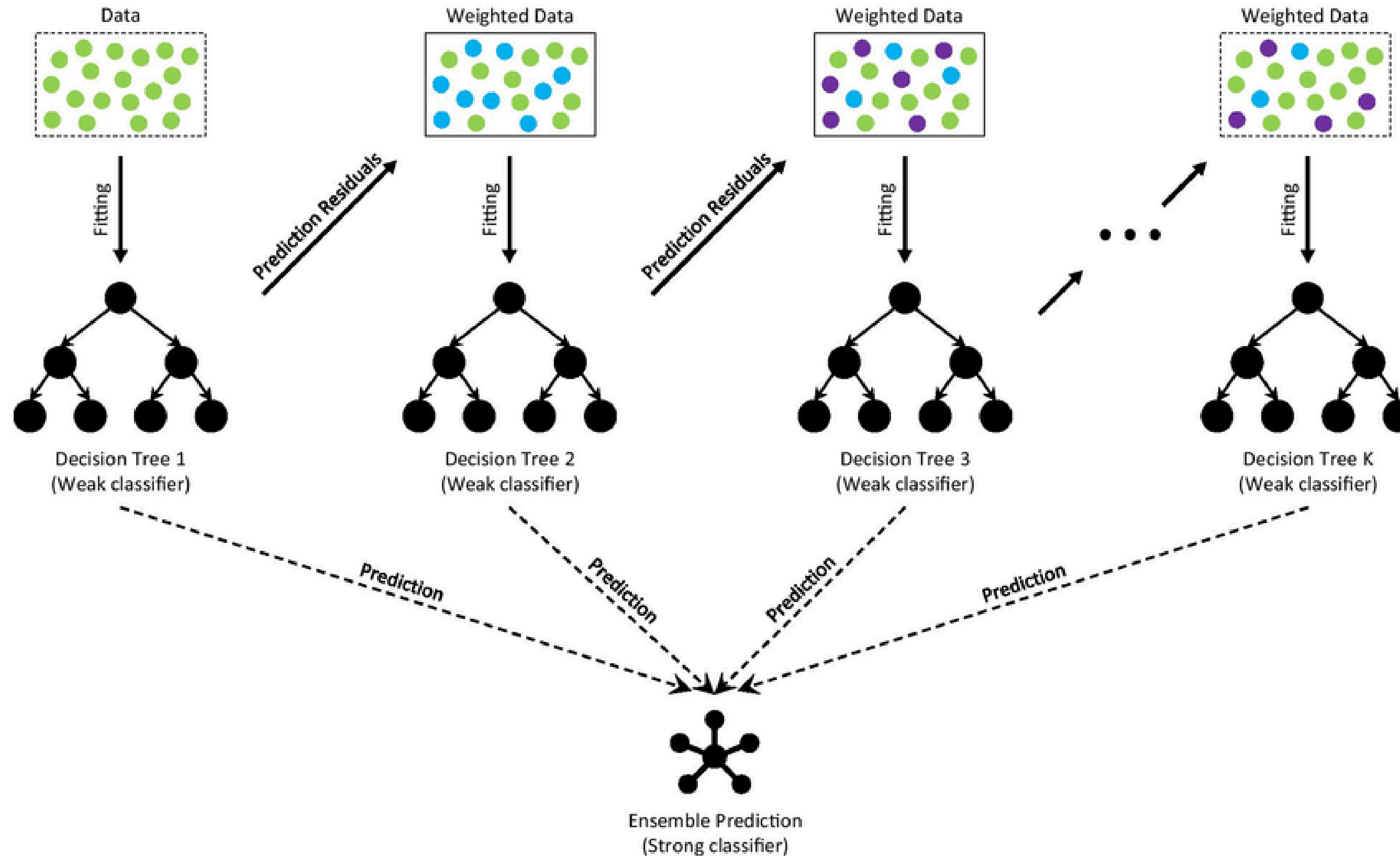
- Ensemble of Decision Trees
- Solves overfitting & high variance
- Uses Bootstrap Aggregation (Bagging)
- Trains trees on bootstrapped data samples
- Final prediction = Average of all tree outputs

RF + Time-Series = Smarter Forecasting

- Works well with high-dimensional data
- Handles macroeconomic forecasting
- Reduces bias & variance
- Learns complex patterns over time
- Used for predicting economic indicators

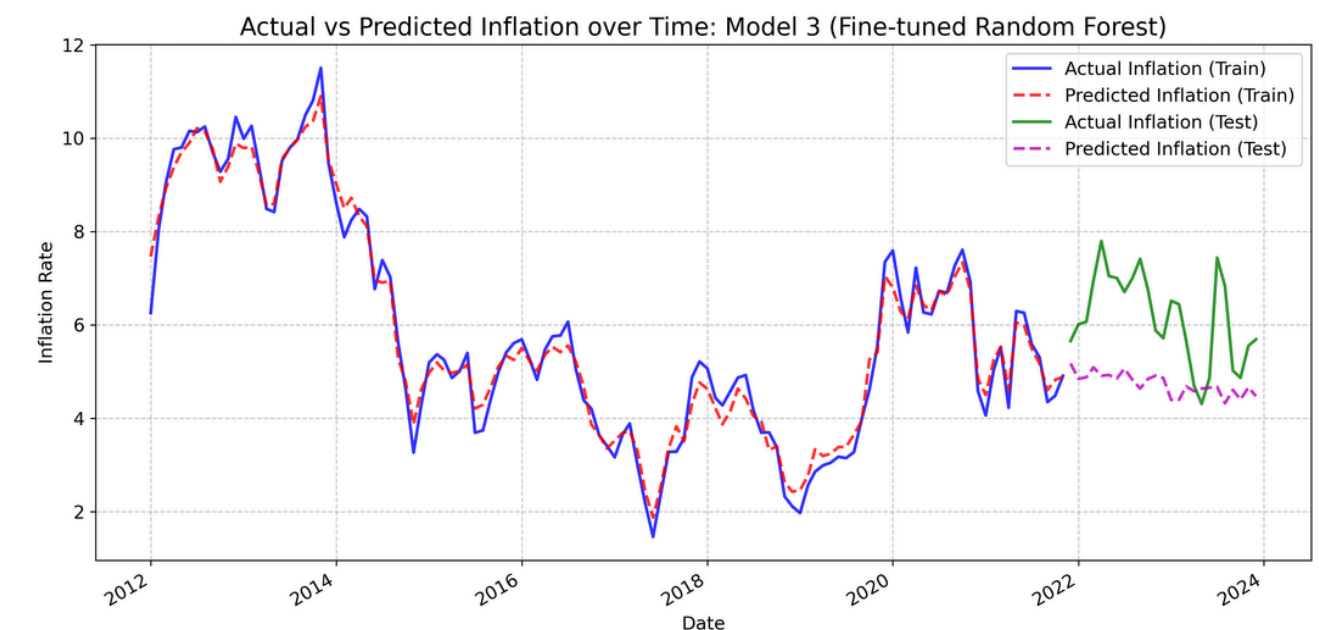
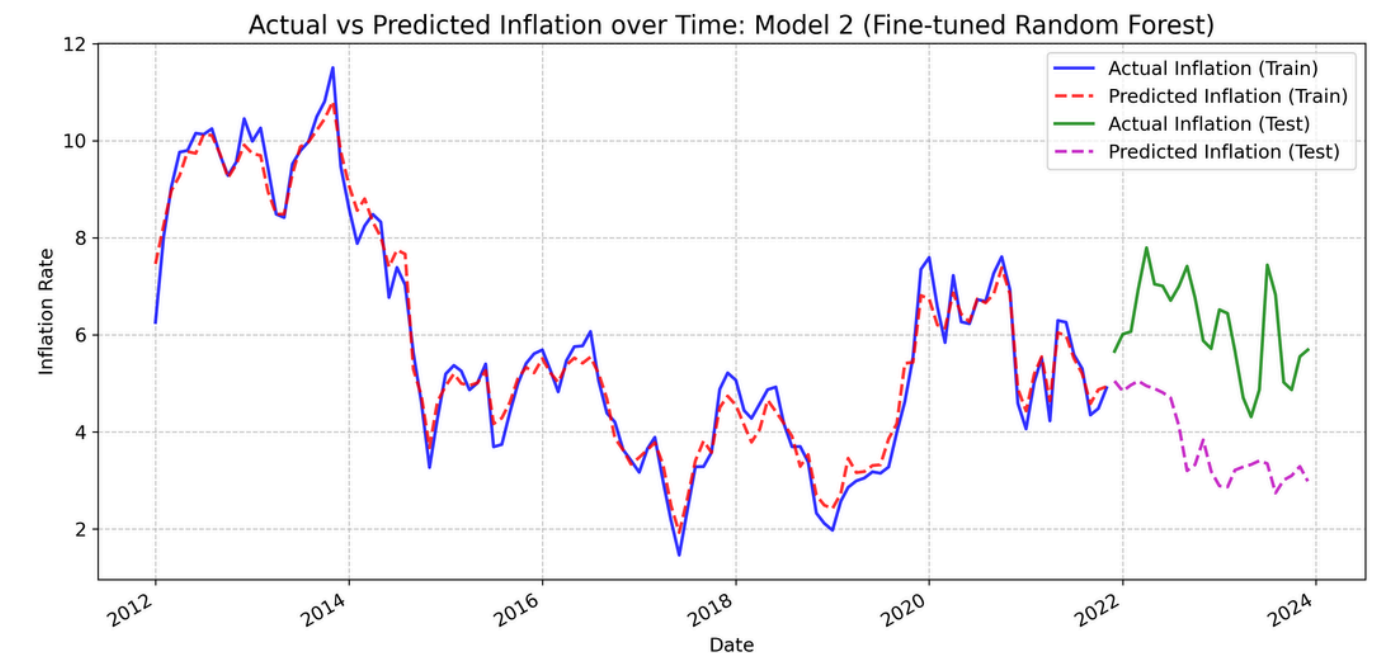
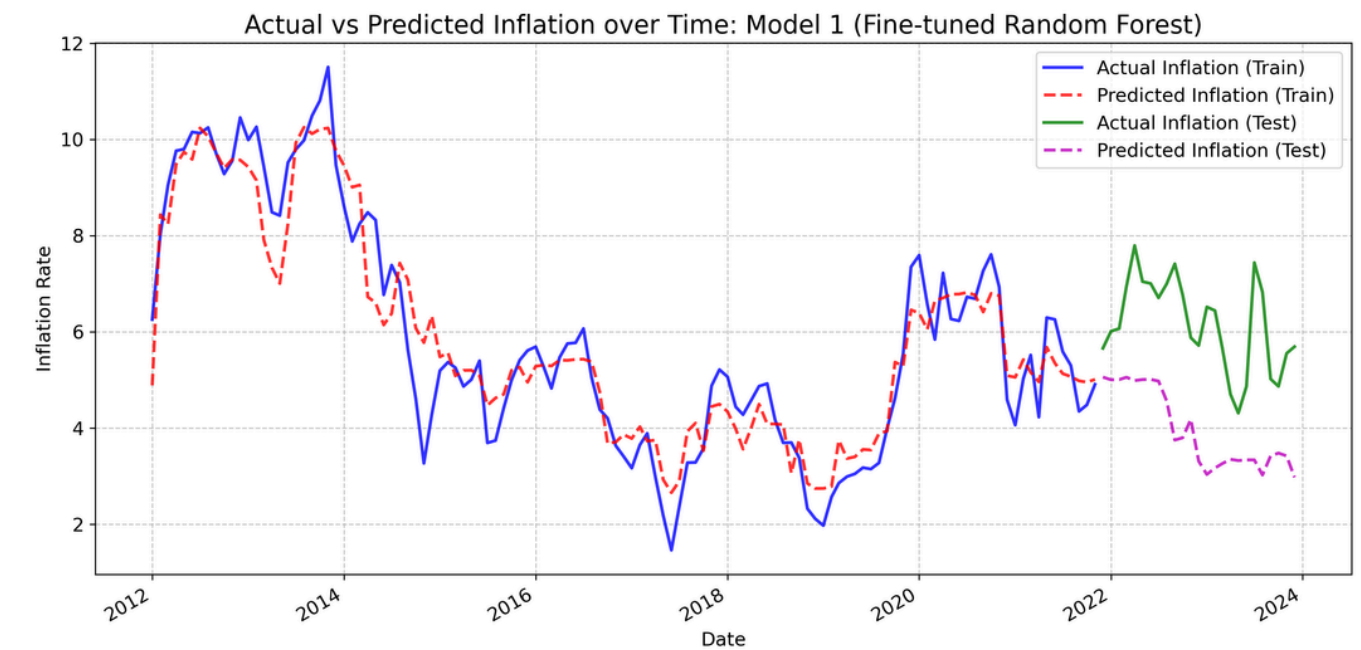


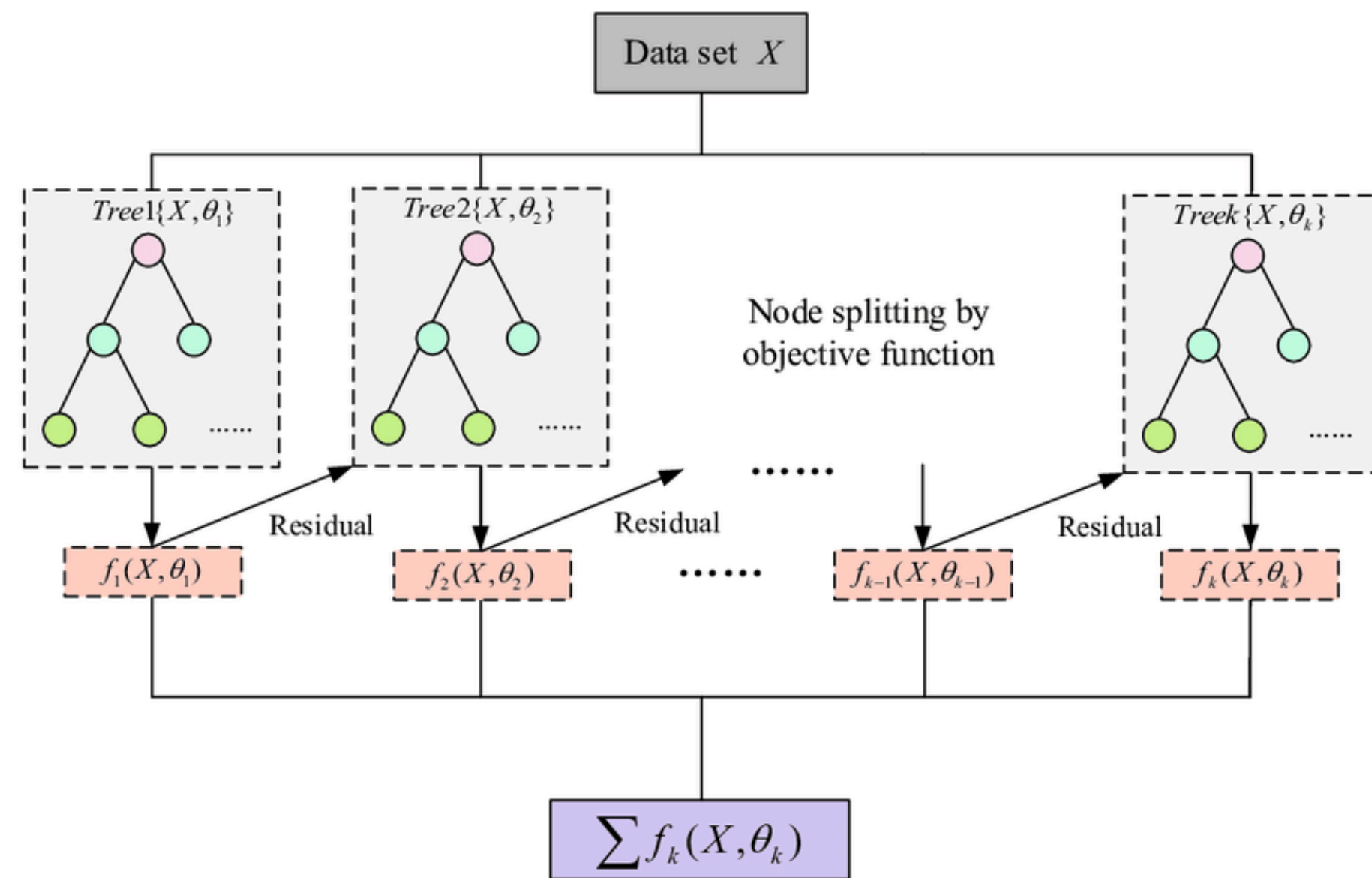
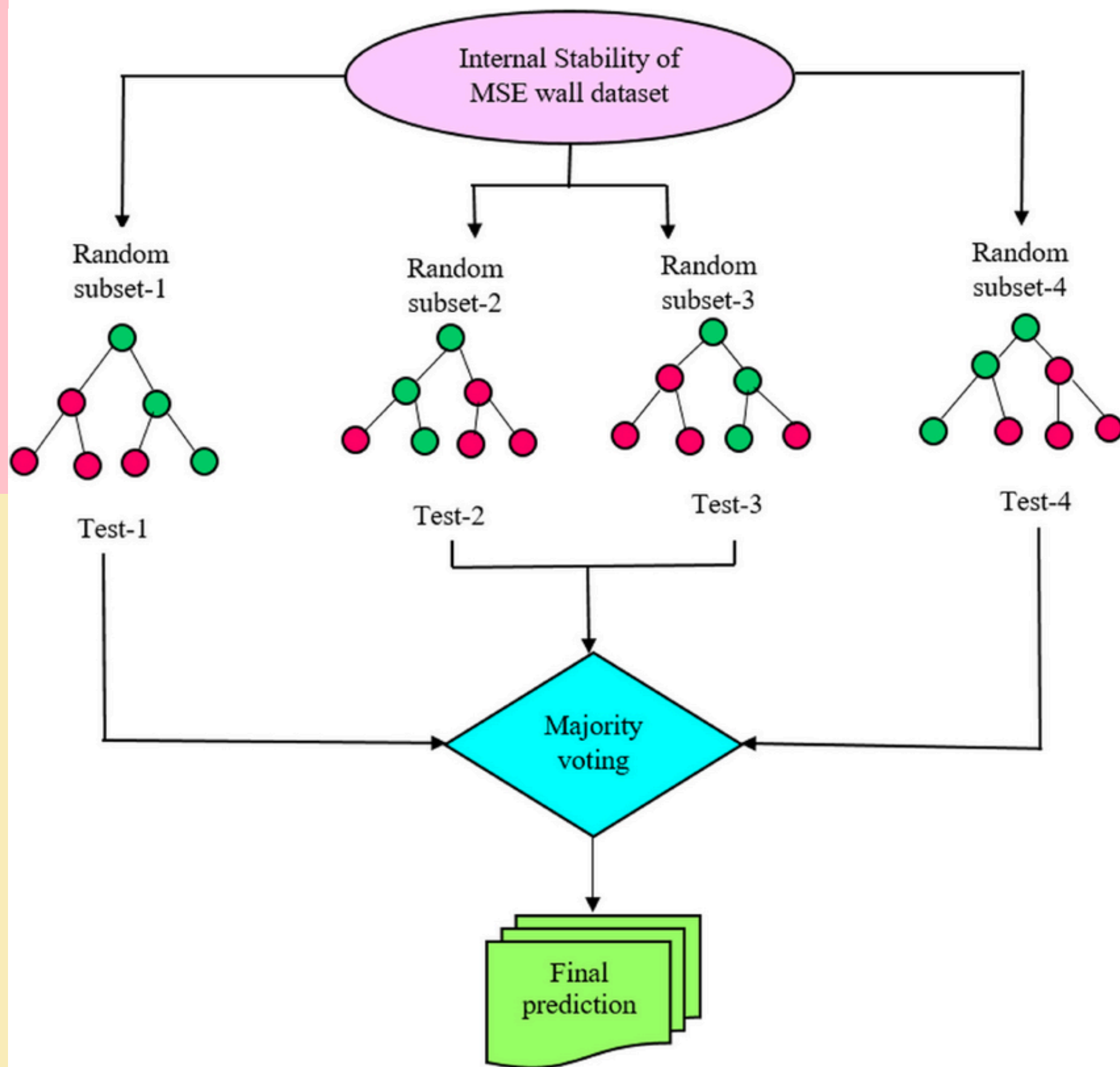
GRADIENT BOOSTING



Why Gradient Boosting Works Well

- Handles non-linear relationships
- Great for high-dimensional data
- Learns complex variable interactions
- Ideal for macroeconomic forecasting
- (Aras & Lisboa, 2022)(Research work)



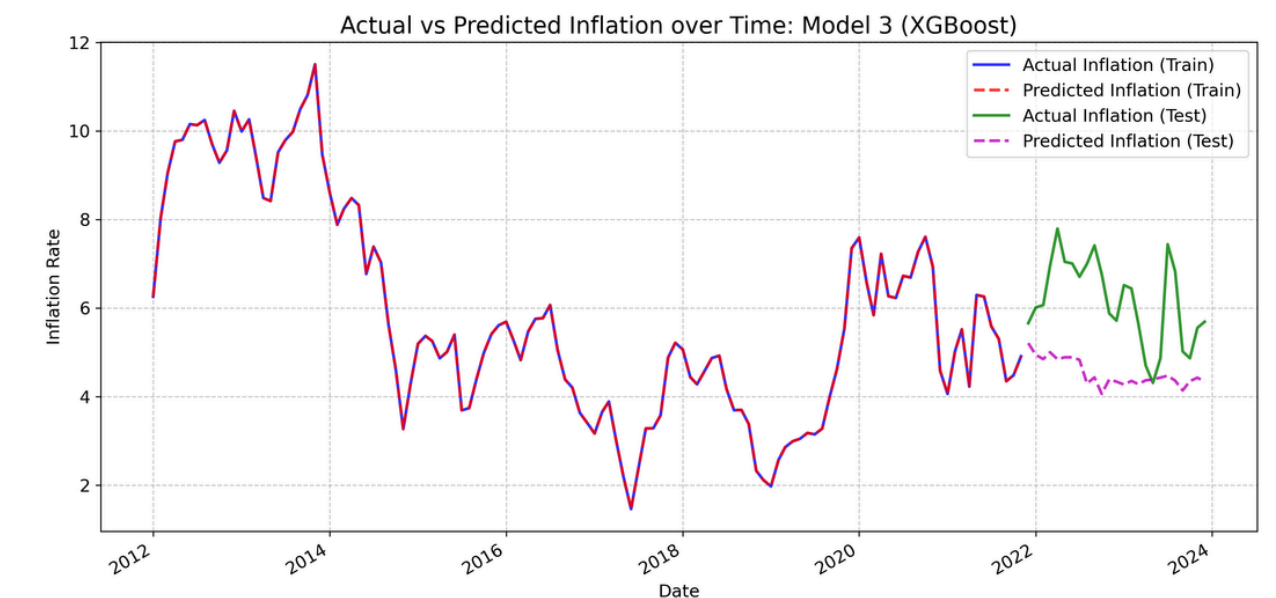
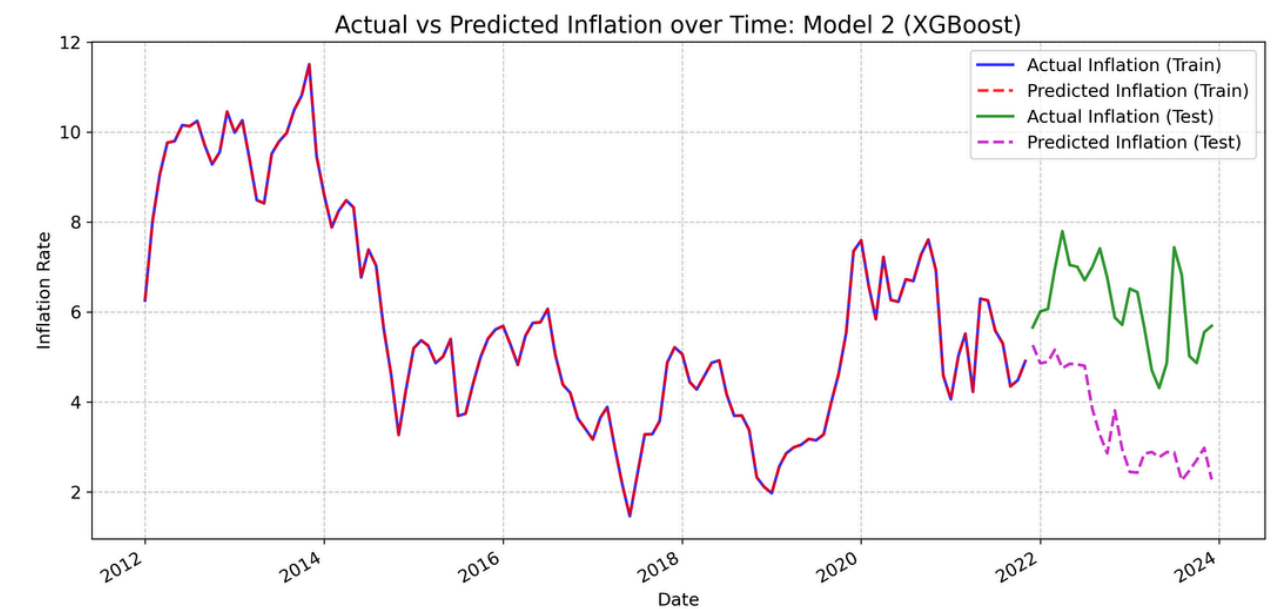
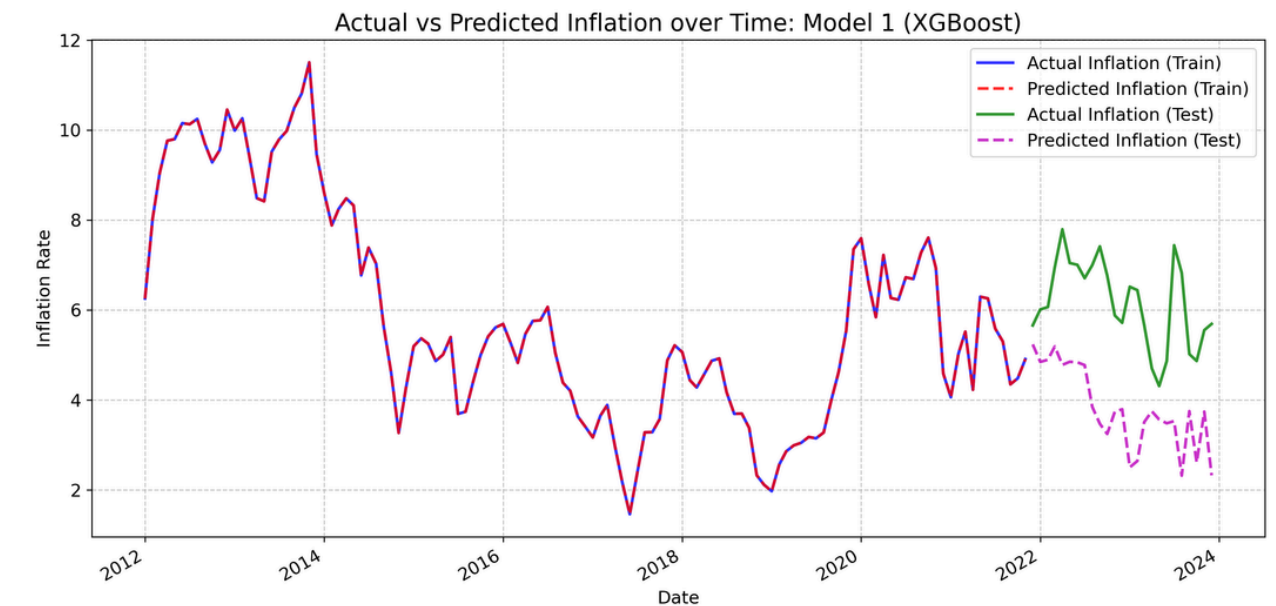


XGBOOST

Why XGBoost Outperforms

- Parallel tree construction → faster training
- Regularization → avoids overfitting
- Custom loss functions
- Excellent accuracy & generalization

XGBoost includes several smart tricks — like training trees in parallel and using L1/L2 regularization. It doesn't just fit the data well; it generalizes better to unseen data. That's why it dominates in data science competitions.



KNN MODEL

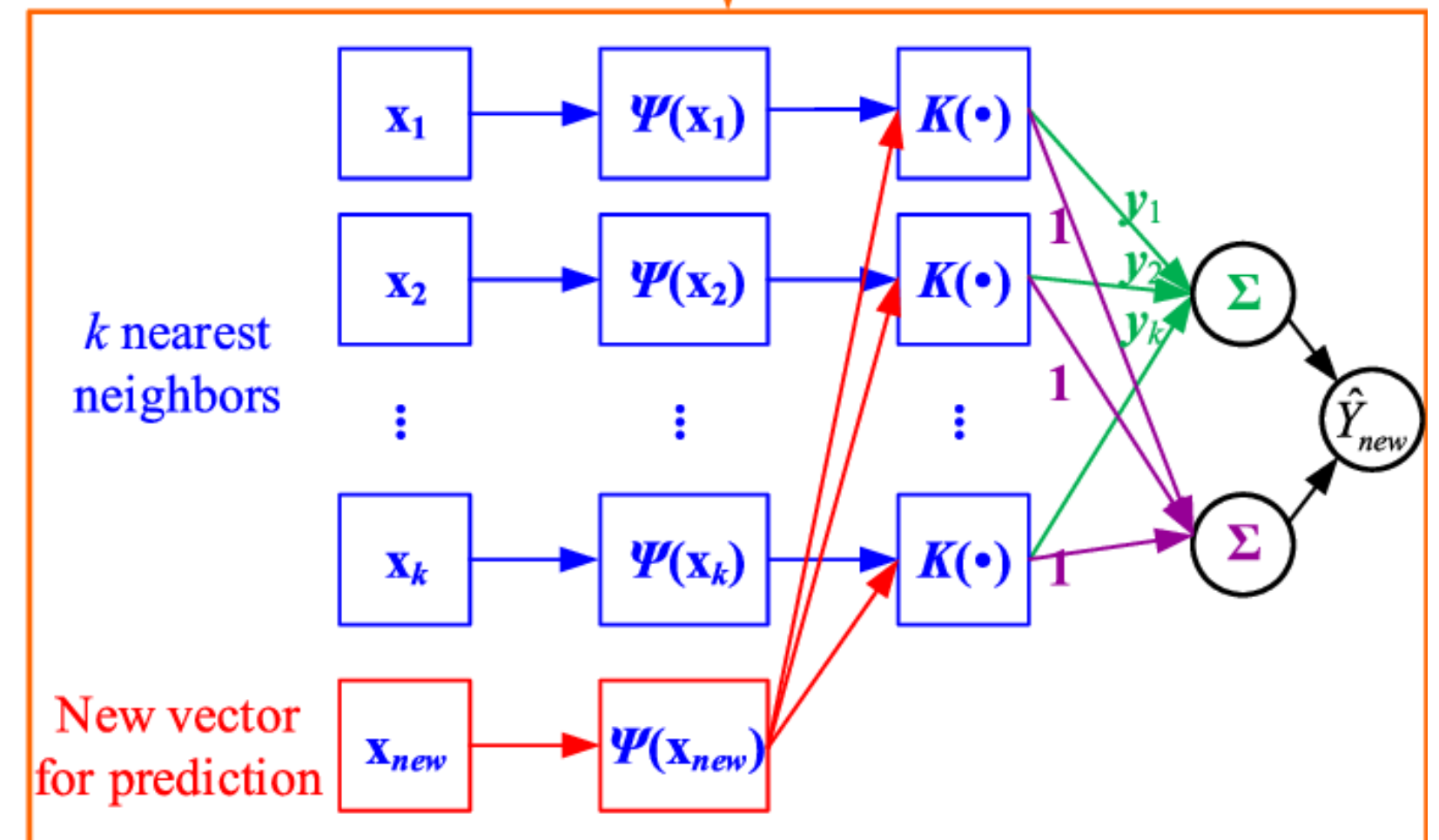
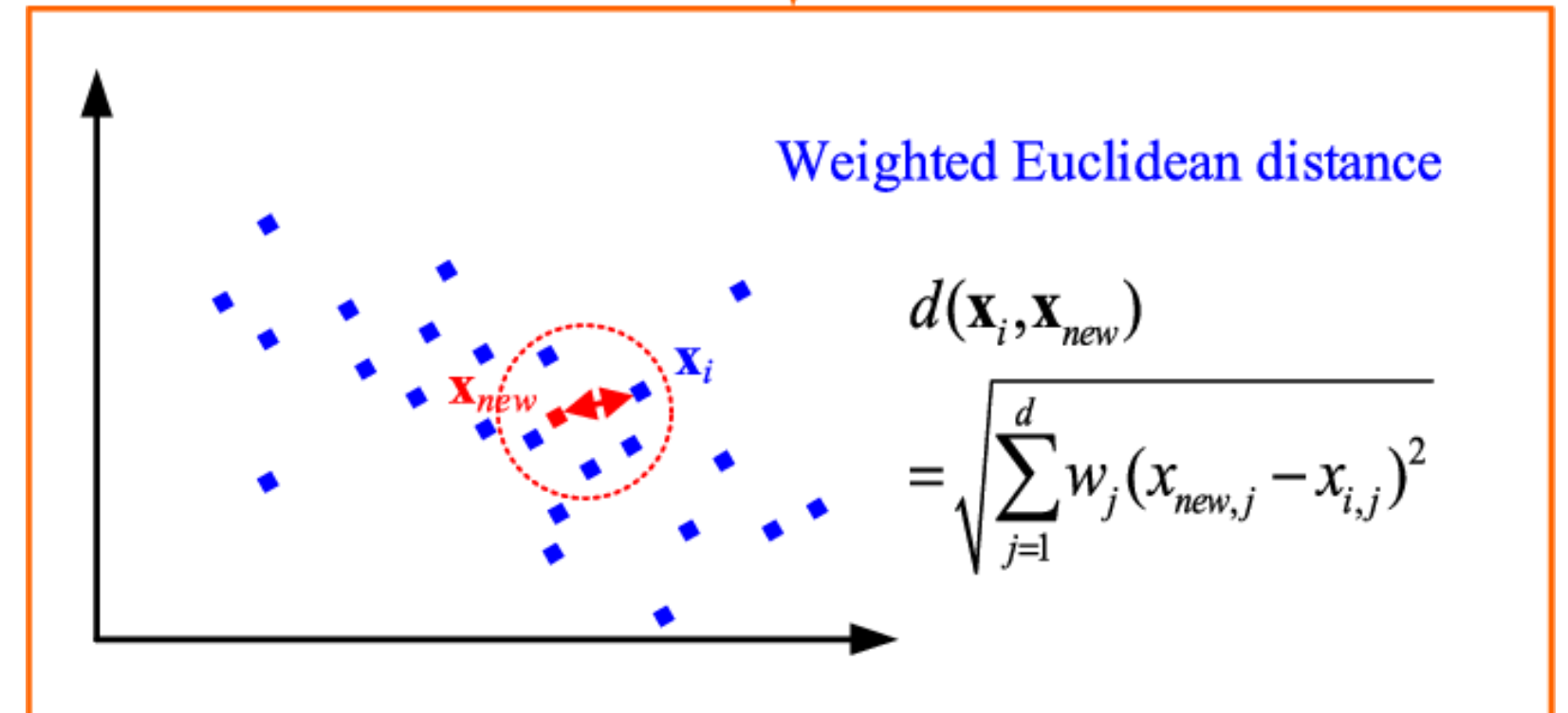
Advantages of K-NN

- Simple and intuitive
- No model training needed
- Can learn non-linear relationships
- Works well with small to medium datasets

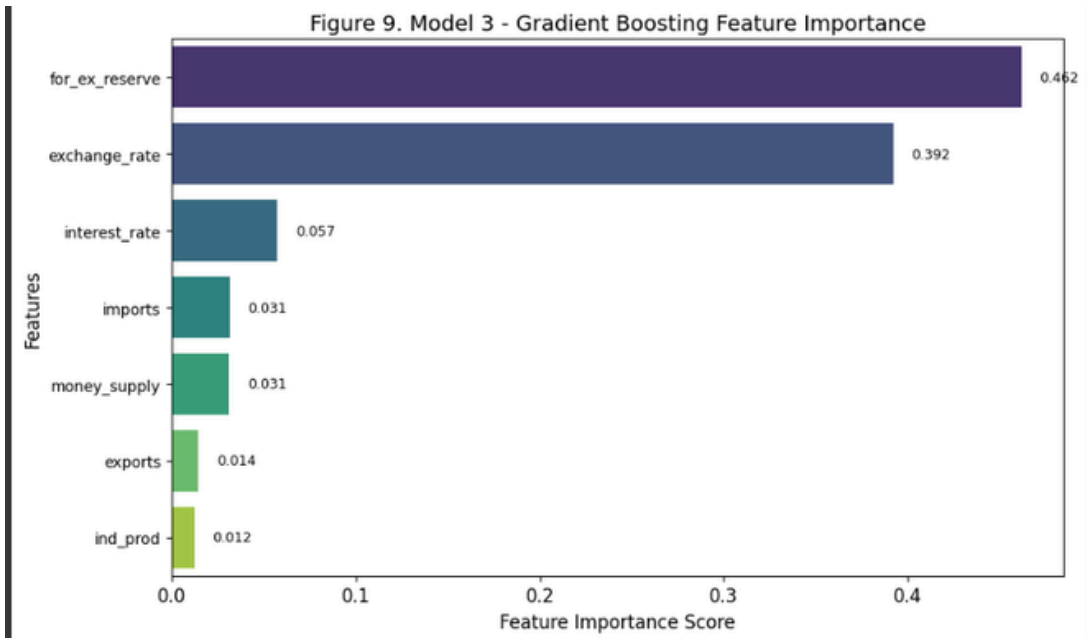
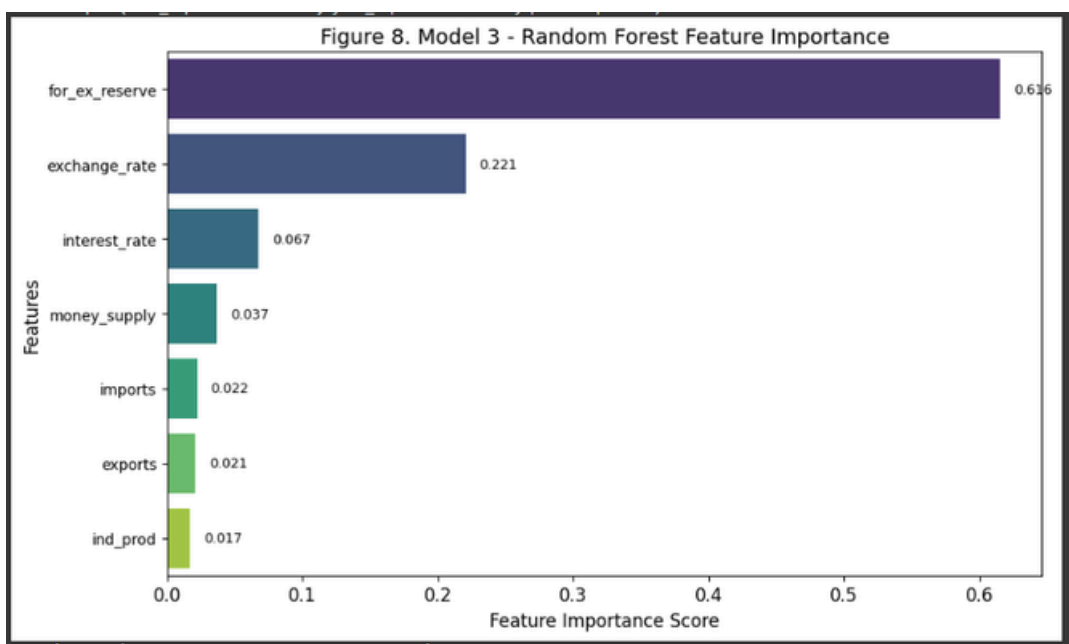
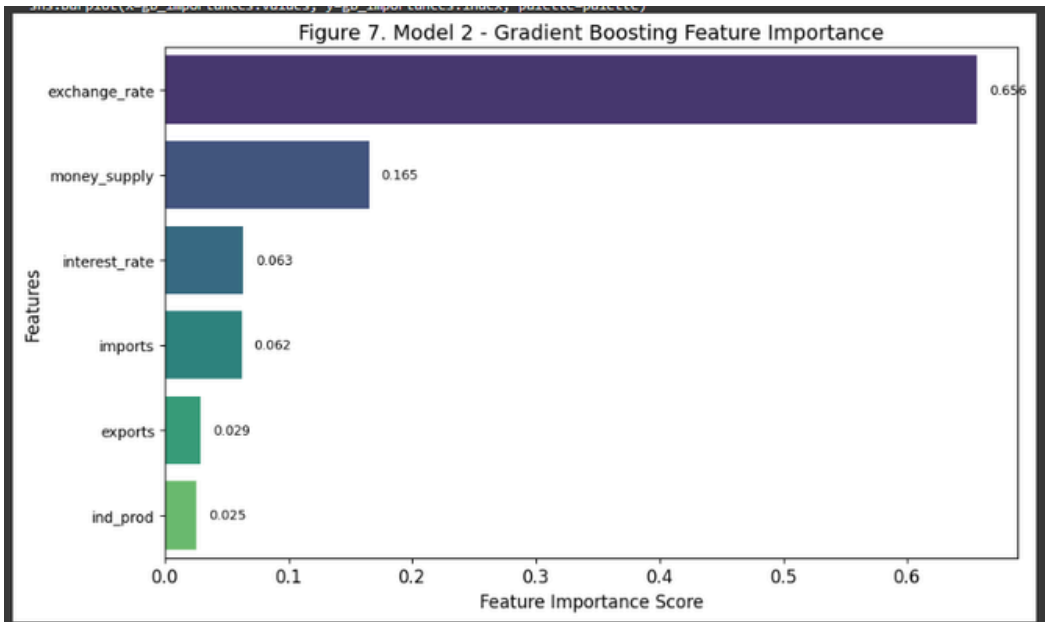
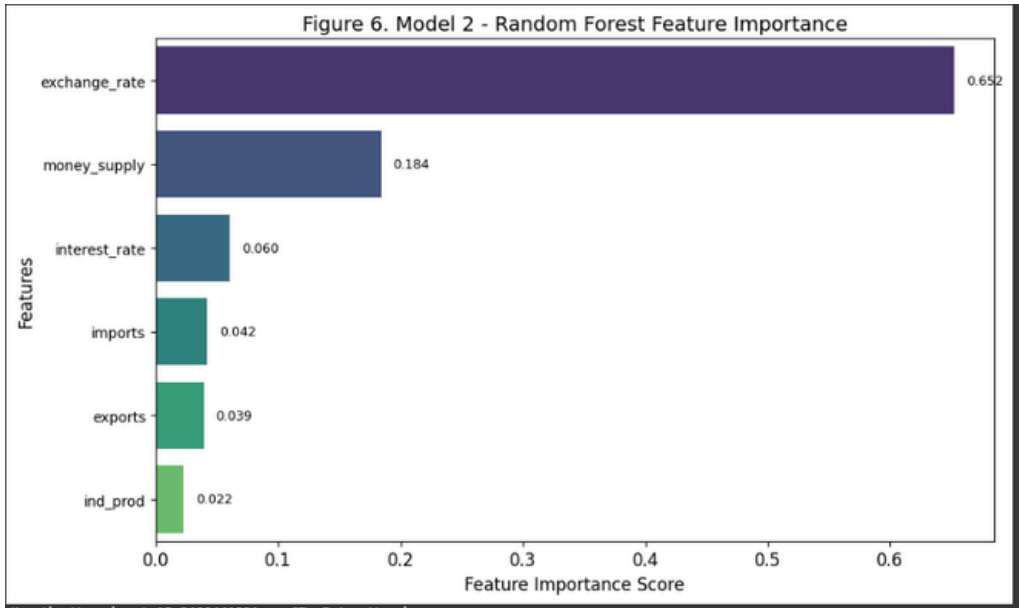
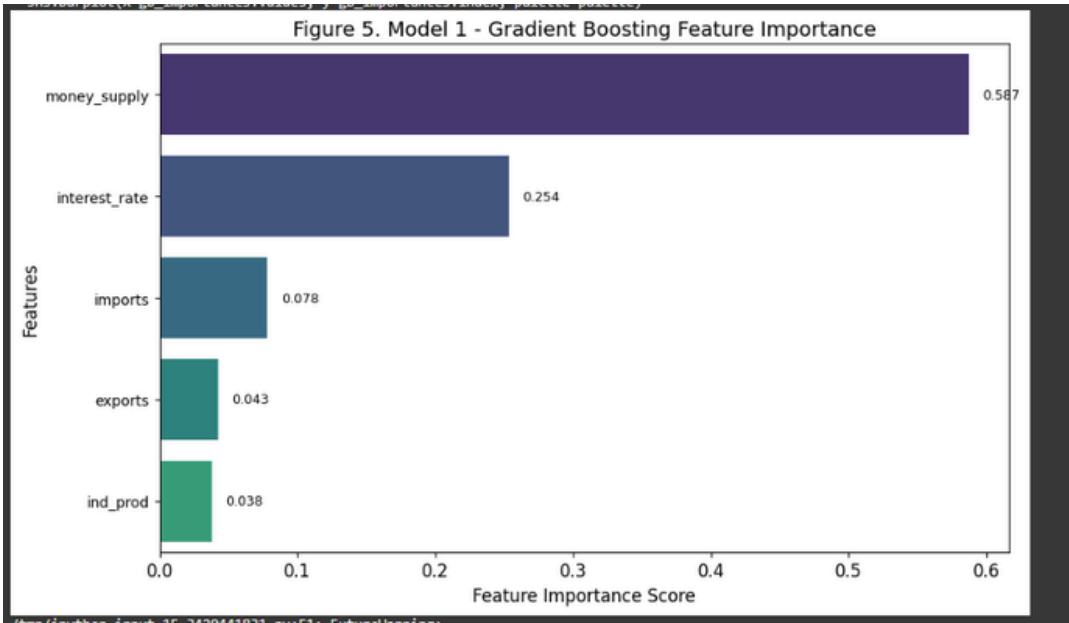
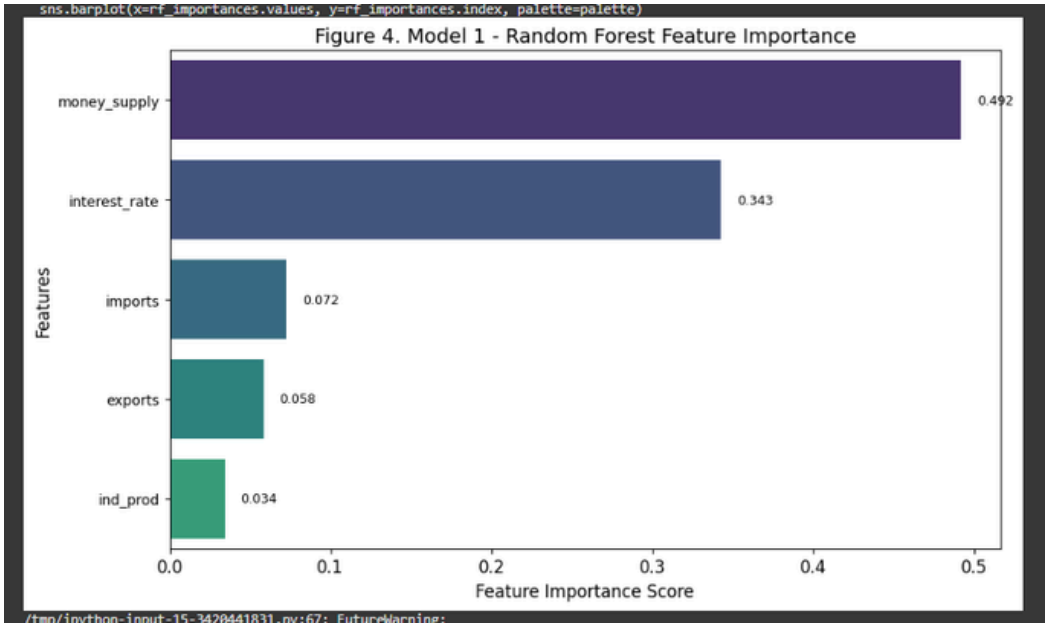
Disadvantages of K-NN

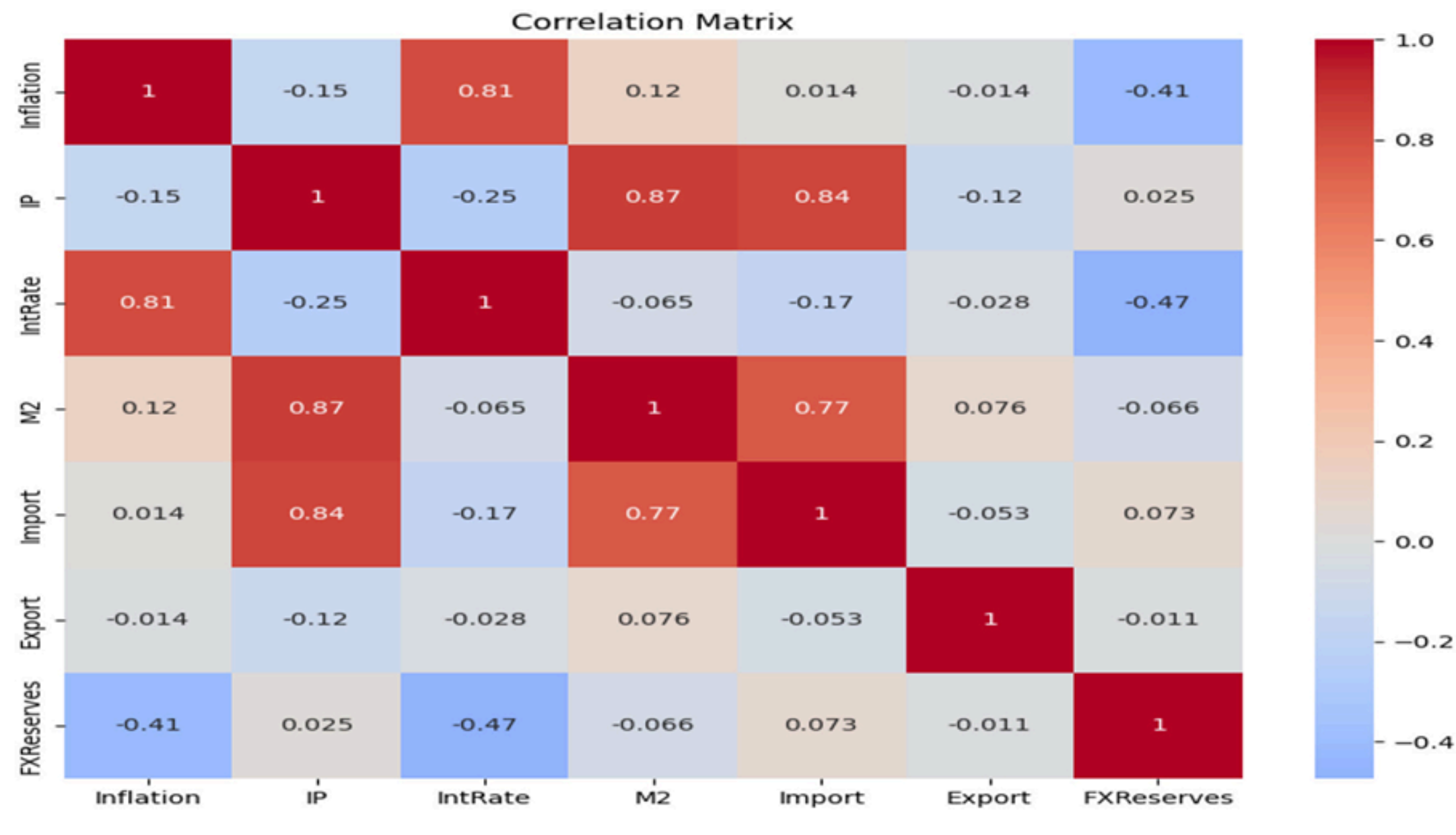
- Slow at prediction (must compute distance to all points)
- Sensitive to irrelevant or scaled features (needs normalization)
- Choice of K is critical:
 - Small K → overfitting
 - Large K → underfitting
- Doesn't handle high-dimensional data well (curse of dimensionality)

$$D_N = \{(\mathbf{x}_i, y_i), i=1, 2, \dots, N\}$$



features_importance





This heatmap shows the pairwise correlation between macroeconomic variables in the dataset (India, 2012–2023). Values range from -1 (perfect negative) to +1 (perfect positive).

- Interest Rate and FXReserves are strongly correlated with inflation → must be included in forecasting models.
- Weak correlation from exports, imports, and M2 suggests their impact on inflation may be indirect or lagged.
- Negative correlation of FXReserves with inflation validates its stabilizing role in emerging economies.

Table 1: Forecasting accuracy.

Panel A: Model 1: A: Model 1

Algorithm	RMSE	MSE	MAPE
Multiple Regression	1.8722	3.5052	0.2674
ARMA	1.022	1.0444	0.1449
ARIMA	1.0782	1.1626	0.1483
Random Forest	2.1149	4.4729	0.2971
Fine-tuned Random Forest	2.1272	4.5251	0.299
SVR	1.1524	1.3281	0.1488
K Neighbourhood	2.0771	4.3144	0.2863
Gradient Boosting	2.5216	6.3582	0.3633
Fine-tuned Gradient Boosting	2.212	4.8928	0.309
XGBoost	2.4349	5.9289	0.3347

Panel B: Model 2: B: Model 2

Algorithm	RMSE	MSE	MAPE
Multiple Regression	2.869	8.231	0.3875
ARMA	1.022	1.0444	0.1449
ARIMA	1.0782	1.1626	0.1483
Random Forest	2.2201	4.9288	0.3121
Fine-tuned Random Forest	2.2201	4.9288	0.3121
SVR	1.1368	1.2924	0.1482
K Neighbourhood	2.0781	4.3185	0.287
Gradient Boosting	2.2442	5.0365	0.3252
Fine-tuned Gradient Boosting	2.6043	6.7825	0.3687
XGBoost	2.4633	6.0681	0.3489

Panel C: Model 3: C: Model 3

Algorithm	RMSE	MSE	MAPE
Multiple Regression	2.4822	6.1613	0.3246
ARMA	1.022	1.0444	0.1449
ARIMA	1.0782	1.1626	0.1483
Random Forest	1.0881	1.184	0.1539
Fine-tuned Random Forest	1.1063	1.2238	0.1565
SVR	1.0994	1.2087	0.1474
K Neighbourhood	1.6434	2.7008	0.2186
Gradient Boosting	1.1843	1.4025	0.1647
Fine-tuned Gradient Boosting	1.0158	1.0319	0.1458
XGBoost	1.0558	1.1147	0.149

Empirical Observation

- Model 3 (which includes FXReserves) shows a notable drop in RMSE and MSE, compared to Models 1 & 2.
- Indicates that FXReserves significantly enhance prediction accuracy.

If policymakers ignore FXReserves, they risk:

- Misjudging inflation trends
 - Delayed or inappropriate monetary interventions
 - Reduced credibility in macroeconomic planning
- Machine Learning Models Outperform Traditional Models


```
--- Aggregation: weighted_mean ---  
  
Random Forest (weighted_mean) Performance:  
RMSE: 0.8492  
MAE: 0.6812  
R2: 0.8872  
MSE: 0.7211  
  
--- Aggregation: median ---  
  
Random Forest (median) Performance:  
RMSE: 0.8104  
MAE: 0.6722  
R2: 0.8972  
MSE: 0.6567  
  
--- Aggregation: trimmed_mean ---  
  
Random Forest (trimmed_mean) Performance:  
RMSE: 0.8511  
MAE: 0.6933  
R2: 0.8866  
MSE: 0.7244  
  
--- Aggregation: mode ---  
  
Random Forest (mode) Performance:  
RMSE: 0.9033  
MAE: 0.7312  
R2: 0.8723  
MSE: 0.8160  
  
--- Aggregation: topk_mean ---  
  
Random Forest (topk_mean) Performance:  
RMSE: 0.7994  
MAE: 0.6657  
R2: 0.9000  
MSE: 0.6390
```

ADDITIONAL RESEARCH

These are the result of additional changes we did in the model. We actually added different aggregation techniques in the Randomforest techniques which surprisingly gave us much better result ...but they were not better than XG boost.

Additionally in the research paper nothing was given in refernec to the stationarity of the dataset. So we explored ADF test while using ML models

CONCLUSION

- **Machine Learning Outperforms Traditional Models** : ML models (especially Random Forest and Gradient Boosting) show superior accuracy in forecasting inflation compared to traditional approaches like Multiple Regression and ARMA.
- **FX Reserves Are Crucial for Forecasting** :Including foreign exchange reserves in forecasting models improves predictive accuracy, validating their macroeconomic importance.
- **Traditional Models Have Structural Limitations**: Regression and ARMA models struggle to adapt to nonlinearities, external shocks, and complex interdependencies in inflation drivers.
- **Policy Implication: Data-Driven Decision Making**: Policymakers should adopt advanced forecasting techniques and include FX reserves in macroeconomic modeling for more resilient and proactive monetary policies.

LIMITATIONS

- **Single-Country Study** :Focused only on India (2012–2023); results may not be generalizable to other emerging markets.
- **Lack of Global Variables:** Excludes international macro factors like oil prices, global interest rates, or geopolitical shocks.
- **Black Box Nature of ML:** ML models offer high accuracy but limited interpretability compared to traditional models.
- **Static Historical Dataset:** Forecasting based on historical data — doesn't test real-time adaptability or rolling forecasts.
- **No Stress Testing:** Model performance under extreme economic events or crises not separately analyzed.
- **Limited Economic Evaluation Metrics:** Focused on technical metrics (RMSE, MAPE); economic loss or cost-benefit of forecasting errors not explored.

REFERENCES

Research paper: <https://doi.org/10.1016/j.irfa.2024.103238>

Data: <https://insights.ceicdata.com/Name-your-insight/myseries>

Thank You

<https://github.com/nitishy22/Inflation-prediction-in-emerging-economies-Machine-learning-and-FX-reserves-integration-for-forecast.git>